# Questions about matroids (days 1-2)

## 1 Matroid basics

**Question 1.** *Let $D = (V, A)$ be a DAG (i.e. a directed acyclic graph – a directed graph with no directed cycles). An* out-branching *in $D$ is a subgraph $T$ of $D$ such that the underlying undirected graph of $T$ is a tree, $T$ is spanning (i.e. contains all vertices of $D$), and $T$ has a root $v$ such that all edges of $T$ are directed away from $v$. Assume that $D$ has at least one out-branching. Show that the set of out-branchings of $D$, taken as edge sets, form the bases of a matroid.*

*What is the cleanest description of this structure that you can find?*

For the next question, we investigate a notion of closure operations on matroids. Recall that for a matroid $M = (E, \mathcal{I})$, the *rank* $r_M(X)$ of a set $X \subseteq E$ is the maximum cardinality of an independent subset of $X$.

**Question 2.** *Let $M = (E, \mathcal{I})$ be a matroid and $X \subseteq E$. Define a* closure operation *on $M$ as the exhaustive application of the following rule:*

> *If there is an element $e \in E \setminus X$ such that $r_M(X) = r_M(X + e)$, where $r_M$ is the rank function of $M$, then add $e$ to $X$.*

*A* flat *in $M$ is a set $X$ which is closed under this operation. Similarly to so many other notions on matroids, it is possible to define matroids directly via axioms fulfilled by their closure operation, or their flats.*

*Describe (in non-matroid terminology) what flats are in the following classes of matroids:*

1. *Linear matroids*
2. *Graphic matroids*
3. *Gammoids (this one may be subtle, but interesting)*

**Question 3.** *Verify directly that gammoids are matroids.*

## 2 Representations

**Question 4.** *Let $G = (V, E)$ be an undirected graph and let $A \in \mathbb{Q}^{V \times E}$ be constructed like the usual linear representation of the graphic matroid on $G$, except with the following two variations:*

1. *Instead of, for every edge $e = uv$, setting $a_{e,u}$ and $a_{e,v}$ to $+1$ and $-1$, set them both to $+1$.*
2. *As the previous, except instead we let $a_{e,u}$ and $a_{e,v}$ be indeterminate variables $x_{e,u}$, $x_{e,v}$ (e.g. replace every non-zero in $A$ by a large random number).*

*What are the bases of the resulting matroids?*

That question leads onwards into the topic of *frame matroids*.

**Question 5.** *The Fano matroid $F_7$ (see Fig. 1) is a matroid on ground set $E = \{a, b, c, d, e, f, g\}$ whose circuits are the lines $\{a, d, c\}$, $\{a, f, b\}$, $\{b, e, c\}$, $\{a, g, e\}$, $\{b, g, d\}$, $\{c, g, f\}$ and the circle $\{d, e, f\}$. The non-Fano matroid $F_7^-$ is the matroid of rank 3 where $\{d, e, f\}$ is no longer a circuit. Verify the following two claims from the lecture:*

1. *$F_7$ only has a representation in a field of characteristic 2*
2. *$F_7^-$ only has a representation in a field of characteristic other than 2*

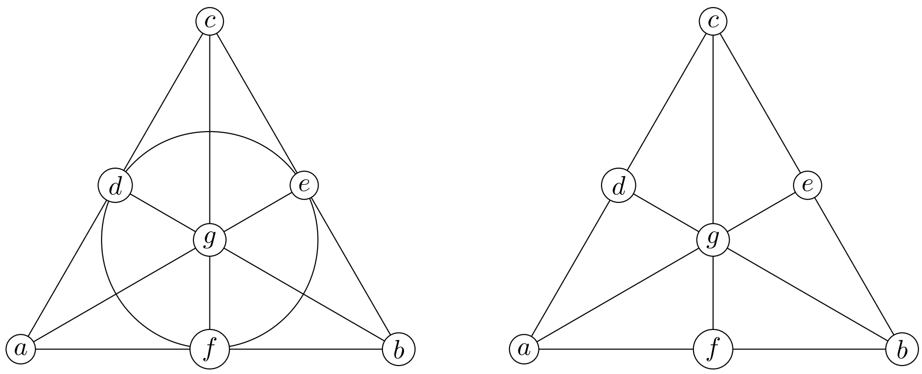


Figure 1: Circuits of the Fano matroid (left) and non-Fano matroid (right)

# 3 Algorithms

**Question 6.** *Verify the correctness of the construction for matroid union.*

**Question 7.** *The problem* FEEDBACK VERTEX SET *is defined as follows: Given a graph $G = (V, E)$, find a minimum set of vertices $X$ whose deletion leaves an acyclic graph. It is NP-hard in general. Show that on instances with maximum degree 3 it reduces to* LINEAR MATROID MATCHING.